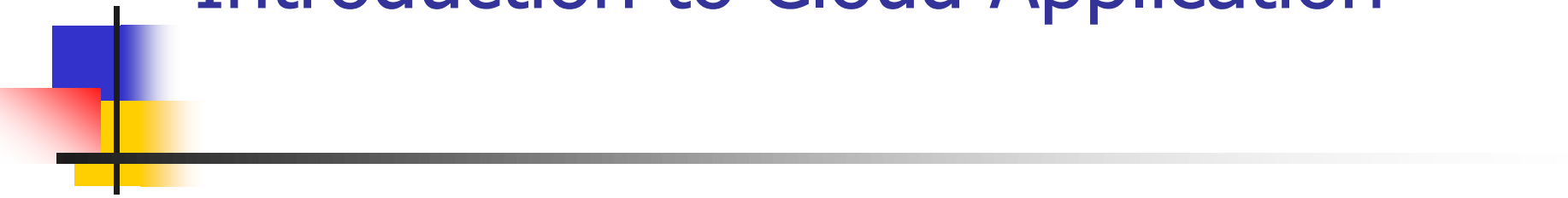
Introduction to Cloud Application



Cloud Application

- A cloud application is software that runs on remote servers and is accessed over the internet, rather than being installed on a local device.

Cloud Application

- Cloud application development involves designing software for cloud environments, and key design considerations include **scalability**, **security**, **performance**, and **reliability**. Other important factors are cost efficiency, manageability, maintainability, and choosing the right technology stack and architectural patterns like microservices.

Cloud Application

- **Scalability**

Design for elasticity, allowing applications to automatically scale resources up or down to handle varying demand. This can be achieved through techniques like stateless architecture and using load balancers.

Cloud Application

- **Security**

Implement security measures from the ground up, such as encrypting sensitive data, using federated identities, and securing network endpoints.

Cloud Application

- **Performance**

Optimize the application based on performance requirements by using caching, read-replication, or load balancing. Metrics like response time and throughput are crucial for evaluation

Cloud Application

- **Reliability and Availability**

Design for fault tolerance to prevent single points of failure, using features like multiple load balancer instances and designing for high availability.

Cloud Application

- **Manageability**

Focus on making the application easy to manage through automated configurations, staged deployments, and observability across complex workflows.

Cloud Application

- **Cost Efficiency**

Design with cost in mind by optimizing resource allocation and using cloud services efficiently. Cloud-native applications often leverage the pay-as-you-go model effectively.

Cloud Application

- **Maintainability and Upgrades**

Structure the application in a way that makes it easy to update and maintain. Loose coupling of components and service-oriented architectures (SOA) facilitate this.

Cloud Application

- **Technology Stack and Architecture**

Choose the right technologies and architectural patterns. Consider stateless and loosely coupled components for better scalability and resilience. Reference architectures and models like SOA, MVC, or RESTful web services can be useful.

Cloud Database

A **Cloud Database** is a **database service delivered through the cloud**. It is hosted and managed by a **cloud service provider (CSP)**, accessible via the Internet.

Cloud Database

- **Key Features**
  - **Scalability**
  - **High Availability**
  - **Managed Services**
  - **Pay-as-you-go model**
  - **Accessibility**

Cloud Database

- **Examples**
  - Amazon RDS (Relational Database Service)
  - Google Cloud SQL
  - Microsoft Azure SQL Database
  - MongoDB Atlas

Cloud Database

- **Types of Data based on Structure**

Type	Description	Examples
Structured Data	Data organized in tables with rows and columns.	SQL databases, financial transactions, sensor readings.
Semi-Structured Data	Has some structure but not strictly tabular.	JSON, XML, logs, social media metadata.
Unstructured Data	No predefined model or structure.	Text documents, images, videos, emails, audio files.

Cloud Database

- **Types of Cloud Databases**

Cloud databases are classified based on **data model**, **deployment type**, and **management approach**.

Cloud Database

- **Based on Data Model**

Type	Description	Examples
Relational Cloud Database (SQL)	Stores structured data in tables using SQL. Ensures ACID properties.	Amazon RDS, Azure SQL Database, Google Cloud SQL
NoSQL Cloud Database	Designed for unstructured or semi-structured data. Scalable and schema-less.	Amazon DynamoDB, MongoDB Atlas, Google Firestore
NewSQL Cloud Database	Combines scalability of NoSQL with ACID compliance of SQL systems.	Google Cloud Spanner, CockroachDB

Cloud Database

- **Based on Deployment Model**

Type	Description	Examples
Public Cloud Database	**Hosted and managed by a third-party provider.**	**Amazon RDS, Azure SQL, MongoDB Atlas**
Private Cloud Database	**Deployed within an organization's private cloud environment for more control.**	**Oracle Private Cloud Appliance**
Hybrid Cloud Database	**Combines both public and private setups for flexibility.**	**IBM Hybrid Cloud, Azure Arc-enabled SQL**

Cloud Database

- **Introduction to NoSQL**

NoSQL (Not Only SQL) databases are non-relational systems designed to store, process, and analyze large volumes of diverse data types (structured, semi-structured, unstructured).

Cloud Database

- **Key Characteristics**
  - Schema-less
  - Scalable
  - High Performance
  - Flexible Data Models

Cloud Database

- **Common Types of NoSQL Databases**

Type	Description	Examples
Key–Value Store	Simple data storage using key-value pairs.	Redis, DynamoDB
Document Store	Stores JSON-like documents.	MongoDB, CouchDB
Column Store	Stores data in columns for analytics.	Cassandra, HBase
Graph Database	Stores data as nodes and relationships.	Neo4j, Amazon Neptune

Cloud Database

- **NoSQL vs SQL**

Feature	SQL Databases	NoSQL Databases
Data Model	Relational (tables, rows, columns)	Non-relational (documents, key-value, graph)
Schema	Fixed schema (schema-on-write)	Dynamic schema (schema-on-read)
Scalability	Vertical (add more power to one server)	Horizontal (add more servers)
Transactions	ACID compliant	BASE (Basically Available, Soft state, Eventually consistent)

Cloud Database

- **NoSQL vs SQL**

Feature	SQL Databases	NoSQL Databases
Query Language	SQL	Varies (e.g., MongoDB Query Language, API-based)
Best Suited For	Structured data, transactional systems	Big data, real-time, unstructured data
Examples	MySQL, PostgreSQL, Oracle	MongoDB, Cassandra, Redis